MILITARY SPECIFICATION

FILTER MATERIAL, LAMINATED PAPER, TEN-PLY

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of laminated paper filter material.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-26 - Bag, Plastic, Polyethylene.

MILITARY

MIL-C-51096 - Core Material, C18.
MIL-B-51097 - Backing Material, C18.

STANDARDS

FEDERAL

FED-STD-123 - Marking for Shipment (Civil Agencies).

US GOVERNMENT PRINTING OFFICE 1981—703-023 5197

FSC 4240

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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temperature range. Hose shall comply with specified requirements

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

DRAWINGS

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

E136-41-600-14 - All Purpose Apparatus, Gas Life Testing, Q95, Assembly.
B136-41-850-1 - Testing Apparatus, Agent, Gas-Sorption, Q216, Assembly.
D136-42-102-5B - Penetrometer, Filter, DOP, Testing, Q127, Assembly.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following publications form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM Standards

D645 - Thickness of Paper and Paper Products.
D828 - Tensile Breaking Strength of Paper and Paperboard.

(Copies may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Manufacture. The filter material shall be a uniform laminated sheet, 26 $\pm$ 1/2 by 21 $\pm$ 1/2 inches, consisting of ten layers of core material conforming to MIL-C-51096, between single outside layers of backing material conforming to MIL-B-51097 except the air flow resistance and DOP smoke penetration shall be such that it meets the requirement of this specification.

3.1.1 Basis weight. Prior to lamination, the basis weight (pounds per 3000 square feet) of the combined core layers, using the basis weight designated on the outside wrap of the core material roll, shall be 2,350 $\pm$ 170.

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3.2 Physical and chemical requirements. The filter material shall conform to table I when tested as specified therein.

TABLE I. Physical and chemical requirements

Property	Requirement		Test paragraph
	Minimum	Maximum	
Airflow resistance, mm water	-	28.0	4.4.4.1
DOP smoke penetration, %	-	0.015	4.4.4.1
Cyanogen chloride (CK), minutes	25.0	-	4.4.4.6
Dimethylmethylphosphonate (DMMP):	-	-	-
As received, minutes	130.0	-	4.4.4.7.1
After aging, minutes	130.0	-	4.4.4.7.2
Thickness, inches	0.450	0.510	4.4.4.2
Tensile strength, lbs/inch			4.4.4.3
Cross machine direction	7.0	-	
Moisture content, %	-	3.0	4.4.4.4
Delamination	No separation	No separation	4.4.4.5

3.3 Preproduction sample. Prior to the start of regular production and following any change in regular production methods, materials, or equipment, a preproduction sample of filter material shall be produced in accordance with this specification for examination and tests (see 4.3).

3.4 Workmanship. The filter material shall be free from foreign matter, wrinkles, burn holes, scorching, charring, tears, cracks, holes, thin spots, blisters, or misalignments when examined in accordance with 4.4.3.3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of continuous 1,000 linear feet filter material shall be manufactured using the same methods, materials, and equipment as will be used during regular production to produce 100 sheets, 26 ± 1/2 by 21 ± 1/2 inches and 100 test strips. The test strip shall be full width and no less than 16 inches long. The test strip shall be taken at shears from the continuous run and shall be taken no less than 4 feet and no more than 9 feet apart.

4.3.2 Inspection procedure.

4.3.2.1 For examination and nondestructive tests. Sample sheets of filter material shall be examined in accordance with the classification of defects (see 4.4.3.3) and MIL-STD-105.

4.3.2.2 For destructive tests. Immediately prior to packaging eighty test strips shall be taken randomly from the samples and test specimens shall be prepared from these test strips as described below. Test specimens shall be tested in accordance with table II. Except for the disks which are packaged with sheets as specified in (c) below, all test specimens shall be individually packaged, sealed, and identified immediately after preparation to prevent moisture pick up and degradation. The test specimens shall be packaged in 7 by 7 inch bags conforming to type II, style I, PPP-B-26.

(a) Eighty 5-1/4 inches diameter disk test specimens shall be from 80 test strips (one from each test strip) for all tests except tensile strength and moisture content tests.

(b) For tensile strength test, the test specimens shall be prepared as specified in ASTM D828 from randomly selected ten test strips (one from each test strip) which were used in (a) above.

(c) For moisture content test, 200 test specimens, 5-1/4 inches diameter disks, shall be died out from the remainder of the test strips and packaged ten disks each with five sheets in accordance with 5.1. Select randomly five packages and remove two disks from each package. These disks shall be tested for moisture content. After test, the disks shall be repackaged and resealed with the original package as specified in 5.1.

4.3.3 Acceptance/rejection criteria. The preproduction sample shall meet the examination and tests specified by 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

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TABLE II. For destructive tests

No.	Characteristic	Sample	Requirement		Acc/Rej Criteria: Acc	Rej	Remarks
1	Air flow resistance	50	Table I	4.4.4.1	0	1	
2	DOP smoke penetration	50	Table I	4.4.4.1	0	1	Same specimen as used in 1 above
3	Thickness	10	Table I	4.4.4.2	0	1	Same specimen as used in 2 above
4	Delamination	10	Table I	4.4.4.5	0	1	Same specimen as used in 2 above
5	Tensile strength	10	Table I	4.4.4.3	0	1	ASTM D828
6	Moisture content	10	Table I	4.4.4.4	0	1	2 from each of 5 packages
7	Cyanogen chloride (CK)	10	Table I	4.4.4.6	0	1	Disks only
8	DMMP as received	10	Table I	4.4.4.7.1	0	1	Disks only
9	DMMP after aging	10	Table I	4.4.4.7.2	0	1	Disks only

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the filter material produced during one work shift not exceeding 12 hours, by one manufacturer, under essentially the same manufacturing conditions. No more than two lots of core material, produced by one manufacturer, shall be used in one lot of filter material.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. Eighty test strips (minimum 16 inch length taken at the shears from the continuous run and no less than 40 feet and no more than 50 feet apart) shall be taken at random for testing as specified in 4.3.2.2 except DMMP gas life after aging in table II.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample sheets of filter material shall be examined in accordance with the classification of defects and MIL-STD-105.

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4.4.3.2 For destructive tests. Test specimens from sample test strips shall be prepared and tested as specified in 4.3.2.2 except DMMP gas life after aging in table II.

4.4.3.3 Classification of defects.

(a) Filter material, gas-aerosol, ten-ply.

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Damage (tears, cracks, holes, blisters or thin spots)
102		Foreign matter
103		Burn holes, charring, or scorching
104		Wrinkles
Minor:	AQL	2.5 percent defective
201		Sheet size
202		Charcoal on top or bottom surfaces
203		Misalignment of core material exceeding 1/4 inch

4.4.4 Tests.

4.4.4.1 Airflow resistance and DOP smoke penetration. Airflow resistance and DOP smoke penetration of the specimens shall be determined at a flow velocity of 160 centimeters per minute (16 liters per minute) using the Q127 DOP Filter Testing Penetrometer Assembly (D136-42-102-5B).

4.4.4.2 Thickness. Three thickness readings shall be taken on each test disk in accordance with method C of ASTM D645 except that conditioning of specimens shall be omitted. The thickness of a specimen shall be the average of three readings.

4.4.4.3 Tensile strength. Tensile strength of the test specimens shall be determined in accordance with ASTM D828, except that conditioning of specimens shall be omitted.

4.4.4.4 Moisture content. Each test specimen (5-1/4 inches diameter disk) shall be weighed to the nearest 0.01 gram (wet weight). Place the specimens in a forced air drying oven at 105° to 110°C and a maximum of 20 percent relative humidity until constant dry weight, within 0.01 gram, is reached. Calculate the percent moisture as follows:

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$$\text{Percent moisture content} = \frac{100\ (\text{wet weight} - \text{dry weight})}{\text{Wet Weight}}$$

4.4.4.5 Delamination. The test specimen (5-1/4 inches diameter disk) shall be lifted from a flat surface so that the lower backing layer is not held by the fingers. The test shall be repeated on the inverted disk. The test specimen shall be considered an integral unit if neither the top nor bottom backing layer separates completely from the core material.

4.4.4.6 Cyanogen chloride. The cyanogen chloride gas life of the specimen, after equilibration at a temperature of 80° ± 10°F and a relative humidity of 80 ± 3 percent and corrected to a challenge concentration of exactly 4000 milligrams per cubic meter (mg/m^3) shall be determined using the Q95 Gas Life Testing All Purpose Apparatus Assembly (E136-41-600-14). Cyanogen chloride gas life is defined as the time needed to attain an effluent cyanogen chloride concentration of 8 mg/M^3 when the specimen is challenged as follows:

(a) Agent: Cyanogen chloride (CNCl), technical grade
(b) Concentration: 4000 ± 200 mg/m^3
(c) Flow: 16.0 ± 0.05 liter per minute (l/min), constant
(d) Temperature: 70° to 90°F
(e) Relative humidity: 80 ± 3 percent

4.4.4.7 Dimethylmethylphosphonate (GB simulant).

4.4.4.7.1 As received. The dimethylmethylphosphonate (DMMP) gas life of the specimen, as received, and corrected to a challenge concentration of exactly 3000 mg/m^3 (without prior equilibration), shall be determined using Q216 Agent Gas-Sorption Test Apparatus Assembly (B136-41-850-1). "DMMP gas life" is defined as the time needed to attain an effluent DMMP concentration of 0.04 mg/m^3 when the specimen is challenged as follows:

(a) Agent: DMMP, not less than 96.0 percent pure (see 6.5)
(b) Concentration: 3000 ± 400 mg/m^3
(c) Flow: 16.0 ± 0.05 $l/\overline{min}$, constant
(d) Temperature: 80 ± 10°F
(e) Relative humidity: Not more than 15 percent

4.4.4.7.2 After aging. Aging of the specimens shall be accomplished by storing them in a forced air circulating oven for 120 hours at a temperature of 212° plus 0° or minus 5°F. Prior to aging, the specimens shall be placed on edge on wire racks in the oven for the duration of the aging period. After aging, the specimen shall be tested for DMMP gas life as specified in 4.4.4.7.1.

4.4.5 Acceptance/rejection criteria. The sample shall meet the examination and tests specified by 4.4.3 to be acceptable.

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5. PACKAGING

5.1 Interplant shipment (see 6.3 and 6.6). The filter material shall be packaged, five sheets and ten sample disks (4.3.2.2) per package; and a uniform quantity of packages per shipping container to provide adequate protection from moisture and physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Markings shall be in conformance with FED-STD-123, and then shipping shall include the contractor's name, address, contract number, date of manufacture, and lot number of material if applicable. In addition, the shipping containers shall be marked with "STORE IN COOL, DRY LOCATION" in minimum 1/4 inch characters.

6. NOTES

6.1 Intended use. This filter material is intended for use in filters that are used with explosive ordnance disposal suits.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) The laboratory to which samples are to be sent.
(c) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation before award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the contractor whether or not sample is acceptable.

6.3 Moisture content. Filter material should be packed immediately as it comes off the production line in order to prevent moisture pick-up.

6.4 Marking of filter material. Ink or other marking fluid containing organic solvents should not be used on the filter material as organic solvents penetrate and affect the adsorptive capacity of the material.

6.5 Dimethylmethylphosphonate (DMMP) test reagent. DMMP may be purchased from Mobil Chemical Co., Industrial Chemical Division, 801 E. Main St., P.O. Box 1136, Richmond, VA 23208; or Virginia Chemical Inc., 3340 W. Norfolk Road, Portsmouth, VA 23703.

6.6 Interplant shipment. Packing for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a sub-contractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal, laboratory, or plant.

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6.7 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description, EA-F-1051, dated 25 June 1973.

Custodian:

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Preparing activity:

Army - EA

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